Practice workbook · Compare blood entering and leaving the kidney

Differences between renal artery and renal vein blood, and what urine solutes are dissolved in.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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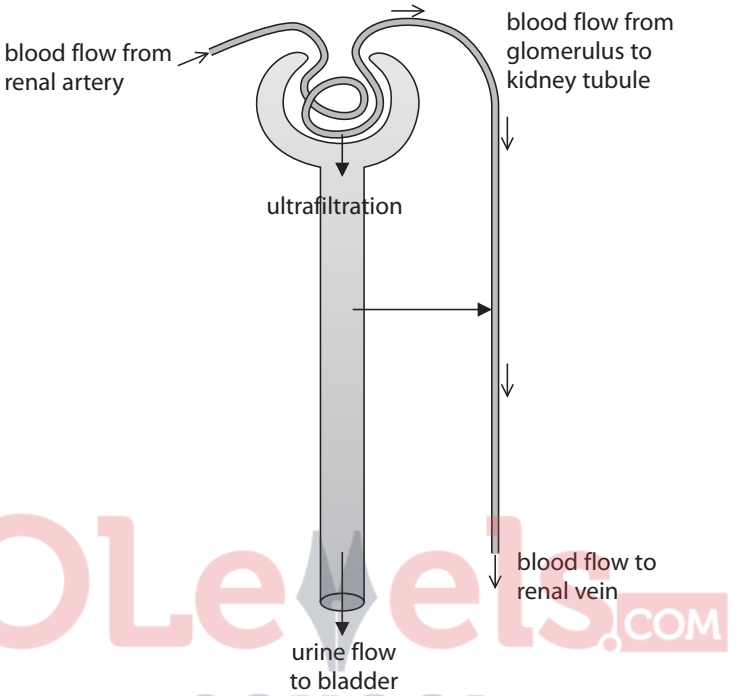
Type ID: HBI-COMPARE-RENAL-BLOOD-COMPOSITION

1. Question 8 · part a.ii · 4 marks

June 2019 Paper 2

Worked-example retry · Attempt part a.ii only.

8 (a) The diagram shows part of a nephron in the human excretory system.



(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

(2)

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1. Question 8 · part a.ii · 4 marks

June 2019 Paper 2

Worked-example retry · Attempt part a.ii only.

(ii) Explain the differences in the composition of the blood flowing through the renal artery and through the renal vein.

(4)

(b) The quantity of salt in the diet affects the concentration of urine produced.

An investigation is set up to test this statement.

This is the method used.

- ten people are each given a different food
- each food contains a different quantity of salt
- the volume of urine produced by each person is recorded each hour for six hours after eating the food

(i) State the independent variable in this investigation.

(1)

(ii) Explain why this investigation is unlikely to produce valid results.

(2)

(iii) State one change to the method that would make the results more valid.

(1)

(Total for Question 8 = 10 marks)

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1. Question 8 · part a.ii · 4 marks · Working and answer

Working and answer

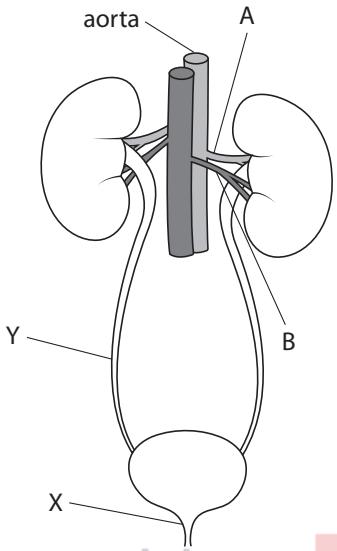
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2. Question 8 · part a.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.iii only.

8 The diagram shows part of the human renal system.



(a) (i) Give the names of structures X and Y. **(2)**

structure X .....

structure Y .....

(ii) Describe the function of the aorta. **(2)**

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(iii) Compare the composition of the blood in vessels A and B. **(2)**

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2. Question 8 · part a.iii · 2 marks

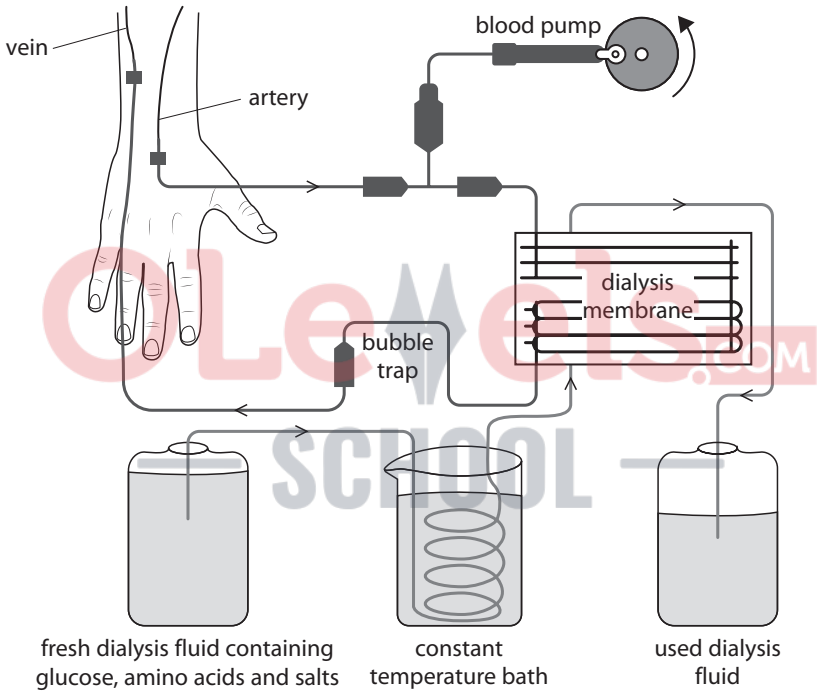
January 2020 Paper 1

Further practice · Attempt part a.iii only.

(b) Dialysis is a treatment used for people with kidney failure.

Blood containing waste substances is pumped through a dialysis machine. The dialysis machine filters the waste from the blood. The cleaned blood is then returned to the patient.

The diagram shows how a dialysis machine filters the blood of a person with kidney failure.



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2. Question 8 · part a.iii · 2 marks · Working and answer

Working and answer

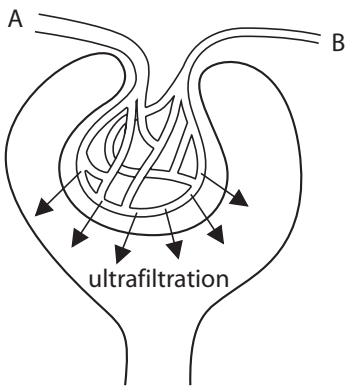
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3. Question 6 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

6 (a) The diagram shows a Bowman's capsule and its associated blood supply.



(i) Describe what is meant by the term **ultrafiltration**.

(2)

(ii) State three differences between the composition of blood in vessel A and the composition of blood in vessel B.

(3)

1 .....

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2 .....

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3 .....

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3. Question 6 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

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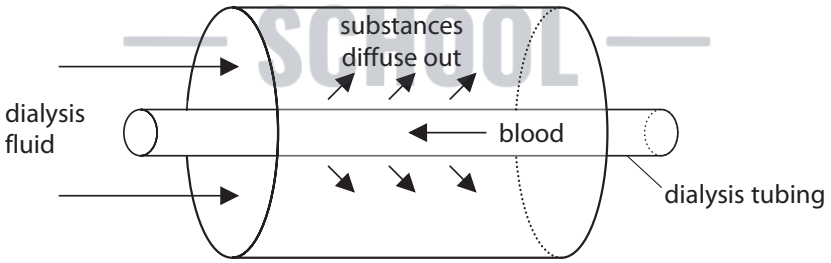
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(iii) Explain how blood vessel A and blood vessel B help the process of ultrafiltration.

(4)

(b) The diagram shows part of a kidney dialysis machine.



The dialysis fluid contains glucose at the same concentration as in the blood flowing through the dialysis tubing.

Suggest the effect on a person undergoing dialysis if the concentration of glucose was below that of the blood flowing through the dialysis tubing.

(3)

(Total for Question 6 = 12 marks)

3. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

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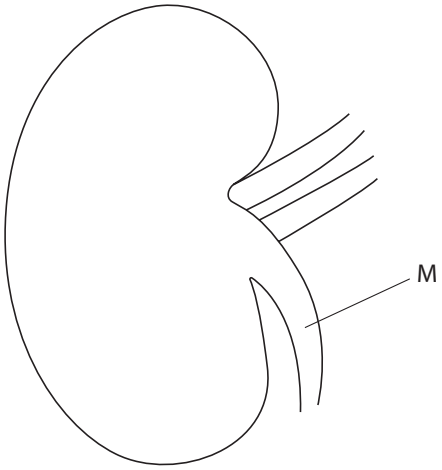
4. Question 7 · part b · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b only.

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7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

4. Question 7 · part b · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

(3)

1 .....

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2 .....

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3 .....

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(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

Substance excreted during 24 hours	High protein diet Mass in g	Low protein diet Mass in g
total nitrogen compounds	16.8	3.6
urea	14.7	2.2
uric acid	0.18	0.09
ammonia	0.49	0.04
creatinine	0.58	0.60

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

(3)

percentage difference = ... %

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4. Question 7 · part b · 3 marks · Working and answer

Working and answer

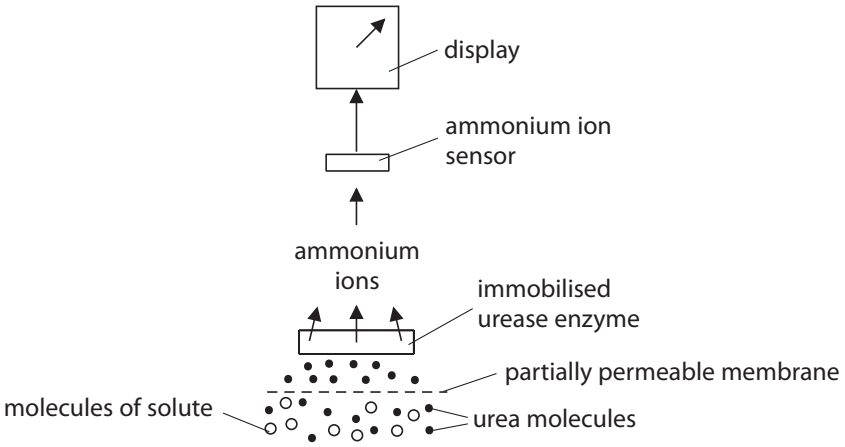
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5. Question 3 · part a.i · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.i only.

3 The diagram shows a biosensor that can be used to measure the concentration of urea in urine.



(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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5. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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